SQL-PYTHON PROJECT - SPOTIFY

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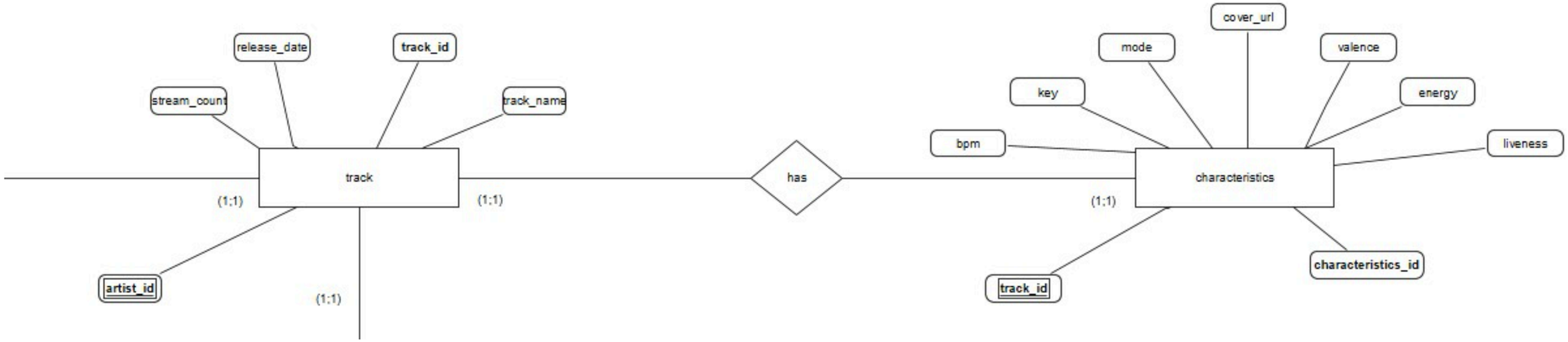
DATASET -

SPOTIFY MOST STREAMED SONGS.CSV

The dataset contains information about the most streamed songs on Spotify. It includes:

- Track and artist names
- Release dates and stream counts
- Audio features (like danceability and instrumentalness)
- Presence in playlists and charts across various platforms
- 950 rows and 25+ columns.

ER DIAGRAM, SCHEMA



DATA_LOADER

Python script to import the dataset into the database using pymysql and pandas.
Steps:

- Cleaned and handled missing data
- Created loops to insert data into each table
- Managed data types and formats (e.g., dates, percentages, commas)

QUERIES

1. Most Instrumental Tracks (from 2000)
2. Most Popular Artists (by stream count)
3. Most Danceable Tracks (after 2020)
4. Tracks per Release Year

```sql
SELECT
    Track.track_name,
    Track.release_date,
    ROUND(Characteristics.instrumentalness / 100, 3) AS instrument
FROM
    Track
JOIN
    Characteristics ON Track.track_id = Characteristics.track_id
WHERE
    Track.release_date >= '2000-01-01'
ORDER BY
    instrumental_score DESC
LIMIT 5;
```

QUERY APP TERMINAL

- Console-based menu
- Simple and intuitive: choose query number, view results
- Works even for users without Python experience
- Each query prints results clearly with labeled rows

```
(venv) (base) giovannilolli@MacBook-Air-di-Luca-4 SQL Project % python 4
.query_app.py

Available Queries:
1. Top Artists by Average Streams per Track
2. Most Popular Artists
3. Most Danceable Tracks After 2020
4. Tracks by Release Year

Enter query number (0 to exit): 1

Results:
track_name: Alien Blues, release_date: 2015-07-24, instrumental_score: 0.910
track_name: METAMORPHOSIS, release_date: 2021-11-25, instrumental_score: 0.900
track_name: Poland, release_date: 2022-06-23, instrumental_score: 0.830
track_name: Forever, release_date: 2019-10-04, instrumental_score: 0.720
track_name: WHERE SHE GOES, release_date: 2023-05-18, instrumental_score: 0.630
```